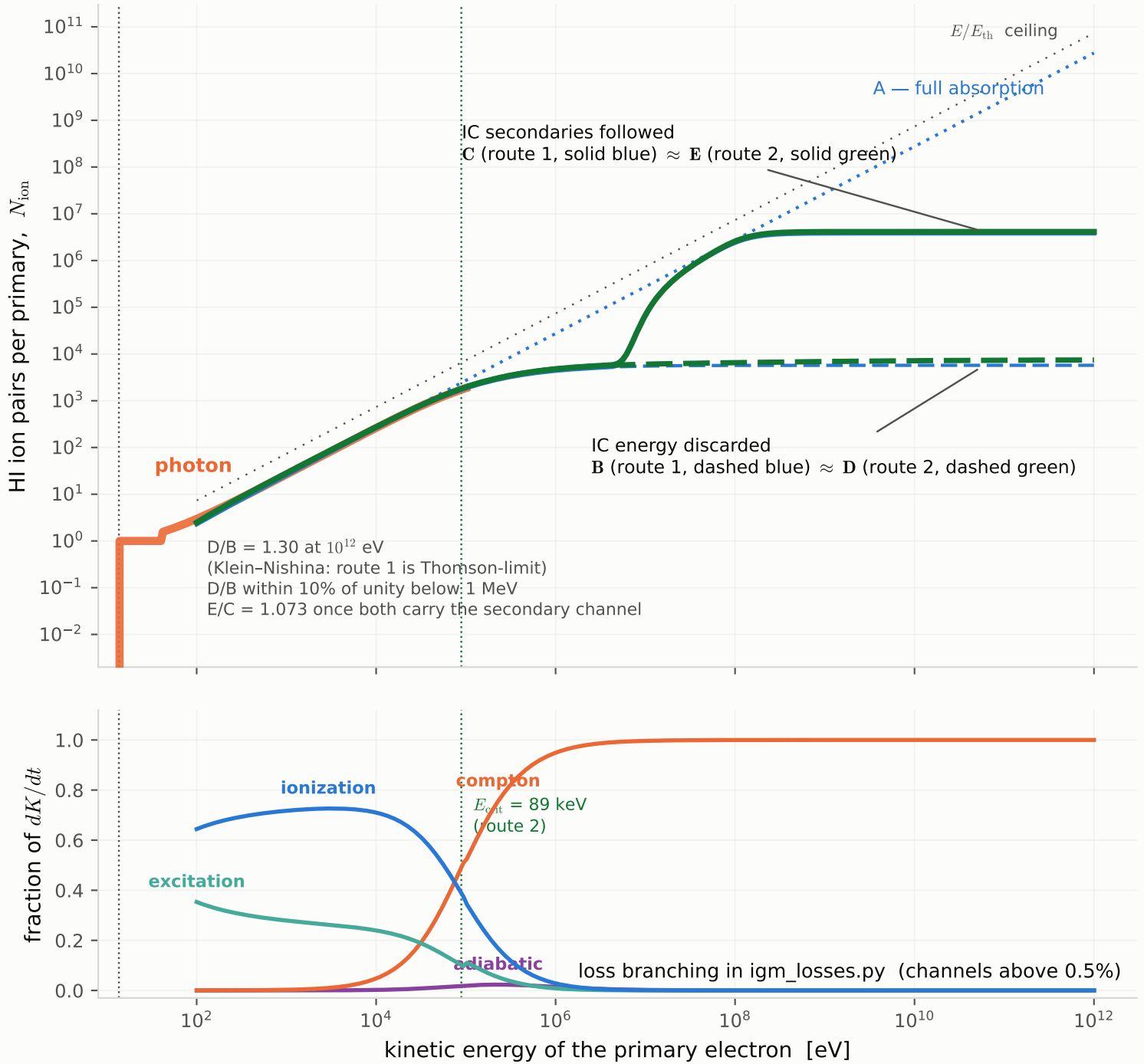


HI ionizations per primary electron: deposition-fit route vs loss-based route IGM at $z = 20$, $x_e = 10^{-4}$, all cascade generations counted



Route 1 (A/B/C): deposition fraction $f_{\text{ion}}(E, x_e)$, Furlanetto & Stoeve 2010 eq. (13); Bethe-Berger-Seltzer stopping; Thomson-limit IC.

Route 2 (D): igm_losses.py — adiabatic, synchrotron, IC with Klein-Nishina, Coulomb (Gould 72), excitation (Stone & Kim 02), ionization (RBED, Kim+00), bremsstrahlung; ionizations counted event by event, knock-on electrons followed through the BED secondary spectrum (Kim & Rudd 94).

Pairings: D is the analogue of B (both discard the IC energy); E is the analogue of C (both follow the upscattered CMB photons).

The curves overlap pairwise. That agreement, between methods sharing nothing but the scenario, is the result.